

Course 01: AI Fundamentals

Comprehensive Learning Guide & Practical Prompting Playbook

MODULE OVERVIEW

Master the core architecture, mechanics, and mental models behind Large Language Models (LLMs). Learn how tokens, context windows, parameters, and temperature affect model output, and set up a foundational framework for predictable, high-quality prompt design.

KEY LEARNING TOPICS

- Understanding LLM Architecture & Tokenization
- Context Windows & Attention Mechanisms
- Temperature, Top-P, and Sampling Parameters
- The Anatomy of an Effective Prompt (Role, Task, Context, Output format)

PROMPTING STRATEGY & ARCHITECTURE

Structure prompts using explicit structural tags (Role, Task, Context, Constraints, Output Format). Define clear parameters for creativity and precision.

ROLE

You are an expert AI Engineer and Computer Science Educator.

TASK

Explain the concept of 'Context Window' and 'Tokenization' to a junior software developer.

CONTEXT

The developer understands basic Python and HTTP APIs but has never worked with LLM architectures or API parameters.

CONSTRAINTS

- Keep the explanation under 300 words.
- Use an analogy related to reading a book or working memory.
- Provide a clear bulleted breakdown of practical implications for API usage.

OUTPUT FORMAT

1. Core Concept (2 sentences)
2. Practical Analogy
3. Key Takeaways for Developers (3 bullet points)

BEST PRACTICES & PRACTICAL TIPS

- Always set explicit constraints on output length, format, and tone.
- Use markdown headers or XML-style tags (,) to separate prompt components cleanly.
- Adjust temperature settings: low (0.0 - 0.3) for factual/code output, higher (0.7+) for creative task generation.